

# Net Productive Entrepreneurship: State and MSA-Level Indices, 2002—2019\*

Justin T. Callais<sup>†</sup>   Peter T. Calcagno<sup>‡</sup>   Gary A. Wagner<sup>§</sup>

## Abstract

This paper extends Sobel’s (2008) entrepreneurship indices, which have over 1,100 citations for testing Baumol’s (1996) theory that institutions incentivize entrepreneurial behavior as either productive or unproductive. Leveraging new datasets, we create annual measures of productive, unproductive, and net productive entrepreneurship for states and metropolitan statistical areas (MSA) from 2002-2019. We document dramatic temporal changes within MSAs, substantial within-state variation across different metro areas, and declining net productive entrepreneurship over time, primarily in populous MSAs. The indices correlate with income, employment, and inequality, and are a valuable resource for exploring empirical questions about entrepreneurship across time and space.

Keywords: productive entrepreneurship, unproductive entrepreneurship, net productive entrepreneurship, index

JEL Codes: L26, M13

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<sup>†</sup>Chief Economist, Archbridge Institute, Email: JustinTCallais@gmail.com.

<sup>‡</sup>Professor of Economics, College of Charleston, Email: calcagnop@cofc.edu.

<sup>§</sup>Acadiana Business Economist Endowed Chair and Professor of Economics, University of Louisiana at Lafayette, Email: gary.wagner@louisiana.edu (corresponding author).

# 1 Introduction

Baumol’s (1996) seminal work describes how institutions shape entrepreneurial talent allocation into productive, unproductive, or destructive activities.<sup>1</sup> Productive entrepreneurship generates value through innovation and growth; unproductive entrepreneurship involves rent-seeking without creating new value; destructive entrepreneurship undermines economic welfare. Crucially, Baumol argues the institutional environment—not entrepreneur supply—determines these allocations. Well-designed institutions (secure property rights, transparent legal systems, competitive markets) channel entrepreneurship productively, while weak institutions incentivize unproductive or destructive activities.

Sobel (2008) conducted the first empirical test of Baumol’s hypotheses by developing a (cross-sectional) net entrepreneurship index for U.S. states in a single year (2000). With over 1,100 citations to date, Sobel’s (2008) index has significantly influenced empirical entrepreneurship research. Utilizing new datasets, we re-envision Sobel’s indices into a panel setting across multiple levels of geography, thereby creating valuable resources for testing empirical questions about institutions and entrepreneurship across time and space.

Landstrom (1999) argues that the roots of entrepreneurship is multidisciplinary including economics, psychology, sociology, and management. Urbano et al. (2019) provides a review of entrepreneurship and institutions and how they affect economic growth, noting that North (1990, 2006), Acemoglu et al. (2014), Baumol (1996), and Rodrik (2003), argue that institutions affect economic growth indirectly. Schumpeter (1949), Leibenstein (1968), Baumol (1996, 1994), and Baumol and Strom (2007) suggest that entrepreneurship affects short and long-term economic growth (Urbano et al., 2019).

We construct productive, unproductive, and net entrepreneurship indices for 50 states and MSAs from 2002-2019, documenting dramatic temporal changes, substantial within-state

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<sup>1</sup>The original foundations appeared in Baumol (1990). Baumol’s (1996) republication has over 10,000 citations.

variation, and declining net productive entrepreneurship in populous MSAs. The indices confirm that productive and unproductive entrepreneurship correlate with economic outcomes including income, employment, and inequality, supporting both Sobel’s (2008) and Baumol’s (1996) findings.

## 2 Data and Index Compilation

We construct our Net Productive Entrepreneurship (NPE) Index following Sobel (2008) but include patent applications as an additional productive measure, since roughly half of applications go unapproved. We also leverage previously unavailable datasets for improved temporal and geographic coverage.

Productive entrepreneurship measures include venture capital expenditures, sole proprietor growth, patents awarded, patent applications, establishment birth rates, and large firm counts.<sup>2</sup>

Employment and establishment data come from the Your Economy Time Series (YETS) database, which captures all U.S. business establishments including owner-only firms, the most inclusive entrepreneurship measure.<sup>3</sup> YETS provides coordinates and six-digit NAICS codes, enabling more nuanced analysis than Sobel (2008).

Venture capital data come from Dow Jones VentureSource, which Andre and Braun (2020) find offers the most accurate coverage.<sup>4</sup>

The unproductive entrepreneurship variables include the number of lobbying establishments, lobbying employment concentration, and State Liability Systems (SLS). We use YETS data to measure lobbying establishment counts and employment within NAICS sector

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<sup>2</sup>Patent applications are unreported before 2002, establishing our start date. PatentsView provides inventor coordinates for MSA-level measurement (Wagner and Bologna Pavlik, 2020).

<sup>3</sup>Government databases require at least one paid employee (Miller et al., 2025), thus undercounting non-employer establishments.

<sup>4</sup>CB Insights acquired VentureSource in 2020.

541820, a classification that encompasses lobbying, political consulting, and public relations establishments.

The State Liability Systems (SLS) survey comes from the U.S. Chamber of Commerce’s Institute for Legal Reform and captures state-level assessments from in-house general counsels, senior litigators, and senior executives at large companies. Respondents evaluate their states across ten dimensions: enforcing meaningful venue requirements; overall treatment of tort and contract litigation; treatment of class action and mass consolidation suits; damages; proportional discovery; scientific and technical evidence; trial judges’ impartiality and competence; juries’ fairness; and quality of appellate review.<sup>5</sup>

For unproductive entrepreneurship, we include lobbying employment concentration—the percentage of the employed population in lobbying—showing relative prevalence compared to other economic activity.

We standardize each of the variables individually using:

$$Standardized_{it} = (Y_{it} - MeanY_i) / (StdDevY_i). \quad (1)$$

Each variable in a given MSA at time  $t$  ( $Y_{it}$ ) is subtracted by the mean and then divided by the standard deviation of the sample. Standardized values are averaged across the productive entrepreneurship variables to obtain a Productive Entrepreneurship Index (PEI). The standardized unproductive variables are averaged to form the Unproductive Entrepreneurship Index (UPI). The NPE is given by:

$$NPE_{it} = PEI_{it} - UEI_{it}. \quad (2)$$

While Sobel (2008) used Borda counting, we standardize variables and compute averages.

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<sup>5</sup>Each state is scored from 0–100. The surveys were conducted in 2002–2008, 2010, 2012, 2015, 2017, and 2019. For missing years, we use linear interpolation.

Standardization preserves relative differences across variables, whereas Borda counting treats rank differences uniformly. Despite outlier sensitivity, standardization avoids the distribution distortions of min-max scaling.<sup>6</sup>

## 3 Results

### 3.1 MSA Index

Table 1 shows top and bottom 10 MSAs by NPE for 2002 and 2019. Several MSAs maintained high rankings (Boulder; San Jose; Dover), while others ranked persistently low (New York; Los Angeles; Tallahassee; San Francisco). Scores declined overall: Hartford’s 2019 10th-place score would have ranked 51st in 2002. Nearly half of top/bottom 10 MSAs in 2002 moved out by 2019, indicating substantial temporal variation.

We illustrate the importance of a time-varying index by reporting the MSAs with the 10 largest changes from 2002 to 2019 in Table 2. Utica-Rome, NY, had the largest increase, moving from 381 to 35. San Jose, CA, had the third largest increase over the sample, although it ranked 2nd in 2002.

Figure 1 illustrates evidence of the spatial variation in NPE within states. Panel A shows the average MSA NPE score, while Panel B shows the change between 2002 and 2019.

In Panel A, California, Michigan, and Pennsylvania each have at least one MSA in all four quartiles. Arizona, Florida, New York, Oregon, Texas, and Washington have at least one MSA in three quartiles. Panel B shows a similar pattern. This highlights the need to examine entrepreneurial activity at a *local* level.

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<sup>6</sup>Under 0-10 scaling, MSAs with 30 and 100 lobbying establishments would score 0.5 and 1.0, while 800 establishments scores 9—creating identical 1-point differences for vastly different gaps.

### 3.2 Correlating Index to Economic Outcomes

To demonstrate the correlation between our indices and economic outcomes, we regress NPE, PEI, and UPI, with MSA and year-fixed effects on real income per capita and employment rates.

In Table 3, we find that NPE and PEI strongly correlate with higher incomes, while UPI correlates with lower real incomes. Increases in NPE are associated with a \$686 increase in income. The effect of the PEI is slightly less than double that.

### 3.3 State Index

Table 4 shows the top and bottom 10 states in the NPE in 2002, 2008, 2014, and 2019. Delaware consistently ranks first, while California always ranks last.

A core premise from Baumol’s argument is that institutions influence the type of entrepreneurship generated. Figure 2 are scatter plots of state-level economic freedom from Stansel et al. (2023) and our NPE, PEI, and UPI indices. We find a weak relationship between economic freedom, NPE, and PEI, but a strong and negative relationship between economic freedom and UPI.

## 4 Conclusion and Discussion

Building upon the theoretical foundations of Sobel (2008) and Baumol (1996), we create state and MSA-level panel indices that allow researchers to investigate both the antecedents of productive entrepreneurship and its consequences for regional economic performance.

Existing studies examine economic freedom’s entrepreneurship effects at multiple levels: international (Bjørnskov and Foss, 2008), state (Wiseman and Young, 2013), spatial (Bologna Pavlik, 2015), and metropolitan (Bennett, 2021). Our indices enable future research to test how individual economic freedom components affect NPE. Research on U.S.

income mobility has concentrated on local social capital (Chetty et al., 2022) and economic freedom (Callais et al., 2025) but has not considered entrepreneurial environments. The NPE index fills this gap, providing researchers with tools to address these and related questions.

Entrepreneurial ecosystems theory argues that entrepreneurship emerges from connections between financial capabilities, human capital, networks, and institutions (Stam, 2015; Spigel, 2017; Autio et al., 2018), with regional differences reflecting ecosystem health. Our index connects to this framework: strong ecosystems emerge from productive entrepreneurship, while weak ecosystems risk channeling effort into unproductive rent-seeking activities lacking institutional support. When institutions weaken conditions for productive entrepreneurship, they degrade the ecosystem’s capacity for innovation and growth. This paper extends entrepreneurial ecosystems theory by demonstrating how productive entrepreneurship shapes whether ecosystems channel talent toward productive or unproductive ends—an important avenue for future research.

## Data Availability Statement

The data and code associated with this version of the paper can be obtained from the corresponding author. The final indices are available at Harvard's Dataverse for other scholars to freely use (<https://doi.org/10.7910/DVN/HFLYDM>).

## **Disclosure of Interest**

The authors confirm that there are no relevant financial or non-financial competing interests to report.

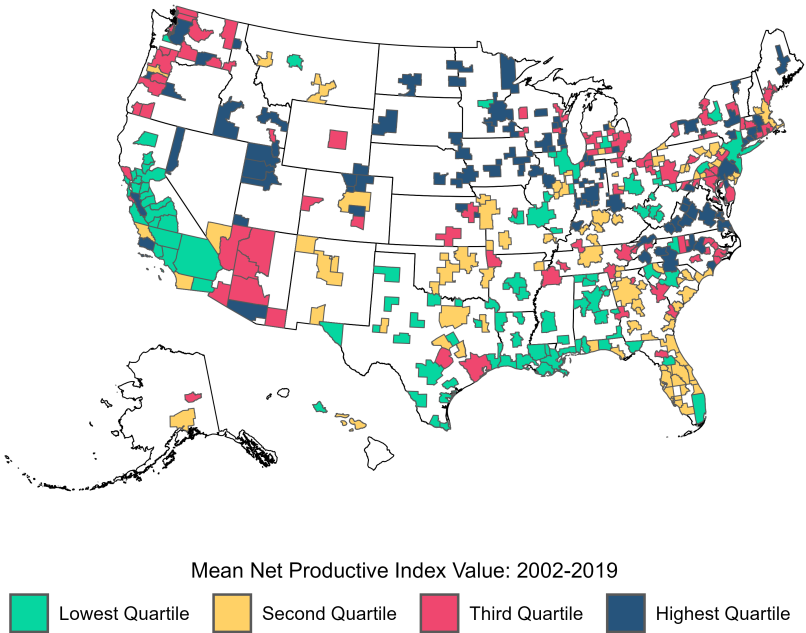
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Figure 1: Net Productive Entrepreneurship by MSA: Average and Change

Panel A



Panel B

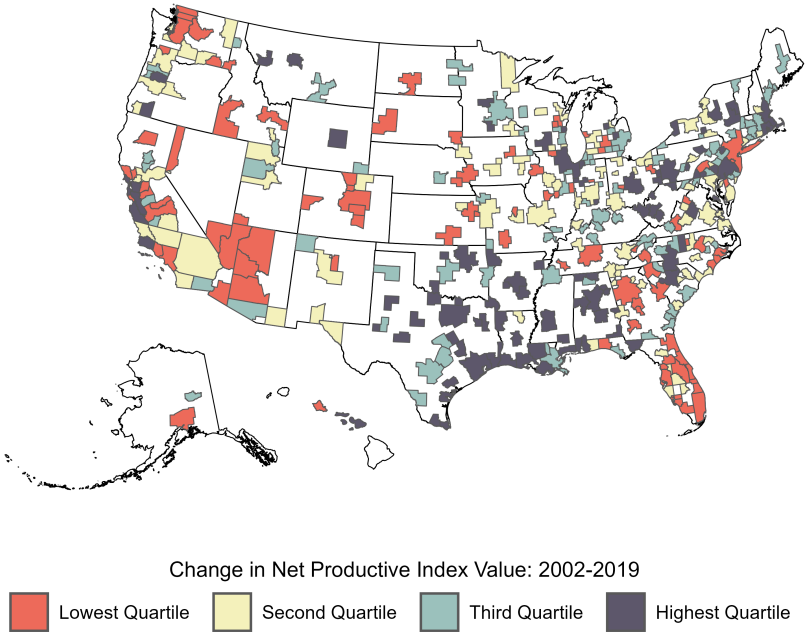


Figure 2: Economic Freedom and NPE

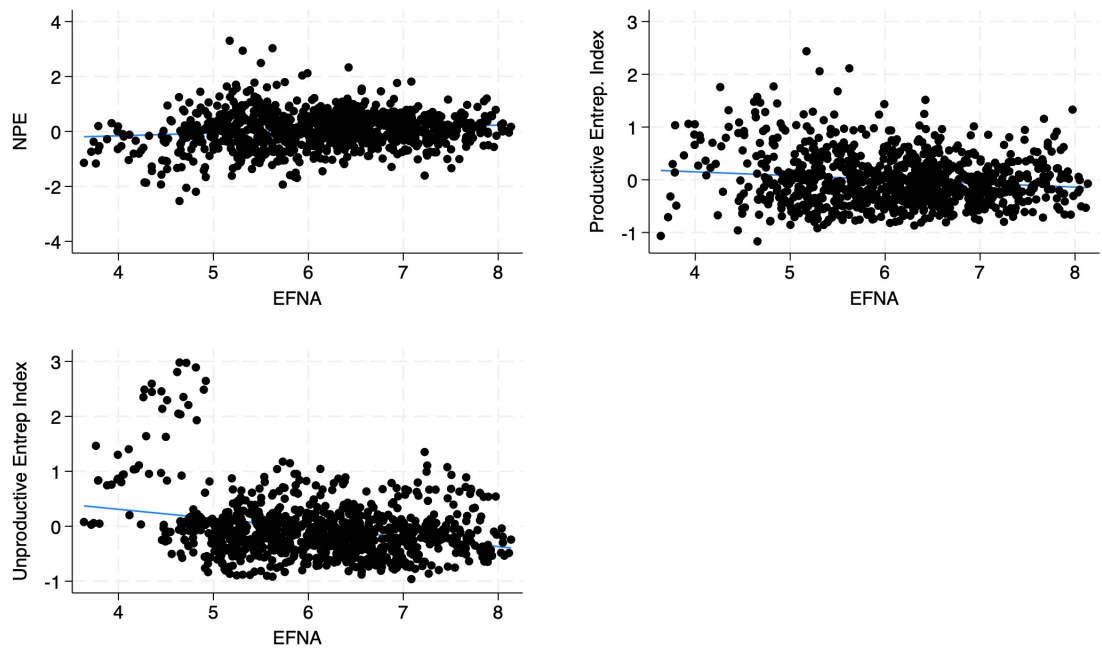


Table 1: Net Productive Entrepreneurship Index Rankings

| 2002 |                                       |        | 2019 |                                           |        |
|------|---------------------------------------|--------|------|-------------------------------------------|--------|
| Rank | MSA                                   | Score  | Rank | MSA                                       | Score  |
| 1    | Boulder, CO                           | 2.346  | 1    | San-Jose-Sunnyvale-Santa Clara, CA        | 2.996  |
| 2    | San Jose-Sunnyvale-Santa Clara, CA    | 1.891  | 2    | Midland, MI                               | 1.444  |
| 3    | Boise City, ID                        | 1.746  | 3    | Boulder, CO                               | 0.911  |
| 4    | Dover, DE                             | 1.615  | 4    | Dover, DE                                 | 0.890  |
| 5    | Oshkosh-Neenah, WI                    | 1.080  | 5    | Columbus, IN                              | 0.831  |
| 6    | Durham-Chapel Hill, NC                | 1.043  | 6    | Corvallis, OR                             | 0.811  |
| 7    | Columbia, MO                          | 1.027  | 7    | Ithaca, NY                                | 0.668  |
| 8    | Wenatchee, WA                         | 1.004  | 8    | Lafayette-West Lafayette, IN              | 0.663  |
| 9    | Omaha-Council Bluffs, NE-IA           | 0.975  | 9    | New Haven-Milford, CT                     | 0.655  |
| 10   | Flagstaff, AZ                         | 0.975  | 10   | Hartford-East Hartford-Middletown, CT     | 0.646  |
| 381  | Utica-Rome, NY                        | -3.592 | 381  | Greenville-Anderson-Mauldin, SC           | -3.677 |
| 380  | St. Cloud, MN                         | -3.116 | 380  | New York-Newark-Jersey City, NY-NJ-PA     | -3.280 |
| 379  | New York-Newark-Jersey City, NY-NJ-PA | -2.907 | 379  | Los Angeles-Long Beach-Anaheim, CA        | -3.120 |
| 378  | Los Angeles-Long Beach-Anaheim, CA    | -2.456 | 378  | Sacramento-Roseville-Arden-Arcade, CA     | -1.631 |
| 377  | Tallahassee, FL                       | -1.872 | 377  | Tallahassee, FL                           | -1.426 |
| 376  | San Francisco-Oakland-Hayward, CA     | -1.613 | 376  | Harrisburg-Carlisle, PA                   | -1.376 |
| 375  | Sacramento-Roseville-Arden, CA        | -1.339 | 375  | Lansing-East Lansing, MI                  | -0.992 |
| 374  | Charleston, WV                        | -1.21  | 374  | Miami-Fort Lauderdale-West Palm Beach, FL | -0.988 |
| 373  | Wheeling, WV-OH                       | -1.084 | 373  | Jefferson City, MO                        | -0.865 |
| 372  | Morgantown, WV                        | -1.084 | 372  | Charleston, WV                            | -0.815 |

Notes: 1 indicates the highest ranking, while 381 indicates the lowest.

Table 2: Largest Increases and Decreases in NPE (2002—2019)

| Largest Decreases               |              | Largest Increases                  |              |
|---------------------------------|--------------|------------------------------------|--------------|
| MSA                             | Change (NPE) | MSA                                | Change (NPE) |
| Greenville-Anderson-Mauldin, SC | -3.637       | Utica-Rome, NY                     | 4.017        |
| Boise City, ID                  | -1.435       | St. Cloud, MN                      | 3.242        |
| Harrisburg-Carlisle, PA         | -1.361       | San Jose-Sunnyvale-Santa Clara, CA | 1.523        |
| Boulder, CO                     | -1.286       | Midland, MI                        | 1.184        |
| Flagstaff, AZ                   | -1.053       | Charlottesville, VA                | 1.137        |
| Cape Coral-Fort Myers, FL       | -1.038       | Enid, OK                           | 1.056        |
| Wenatchee, WA                   | -0.994       | Wheeling, WV-OH                    | 1.012        |
| Seattle-Tacoma-Bellevue, WA     | -0.897       | Gulfport-Biloxi-Pascagoula, MS     | 1.004        |
| Oshkosh-Neenah, WI              | -0.878       | Great Falls, MT                    | 0.942        |
| Lincoln, NE                     | -0.874       | San Francisco-Oakland-Hayward, CA  | 0.932        |

Table 3: Regressing Index on Income per capita

|                                       | <i>Dependent variable:</i> |                        |                     |
|---------------------------------------|----------------------------|------------------------|---------------------|
|                                       | Income per capita          |                        |                     |
| Net Productive Entrepreneurship Index | 686.724***<br>(125.128)    |                        |                     |
| Productive Entrepreneurship Index     |                            | 1243.30***<br>(188.85) |                     |
| Unproductive Entrepreneurship Index   |                            |                        | -254.59<br>(167.97) |
| MSA FE                                | yes                        | yes                    | yes                 |
| Year FE                               | yes                        | yes                    | yes                 |
| R-Squared                             | 0.58                       | 0.60                   | 0.58                |
| Observations                          | 6840                       | 6840                   | 6840                |

Standard errors are clustered at the state dimension. \*\*\* denotes significance at the 1 percent, \*\* at the 5 percent, and \* at the 10 percent level. The regression's constant term, MSA fixed effects, and year fixed effects are not reported.

Table 4: State Scores and Ranks (2002-2019)

| Rank | State | Score (2002) | State | Score (2008) | State | Score (2014) | State | Score (2019) |
|------|-------|--------------|-------|--------------|-------|--------------|-------|--------------|
| 1    | DE    | 1.816        | DE    | 2.122        | DE    | 1.507        | DE    | 1.576        |
| 2    | ID    | 1.215        | OR    | 0.699        | MN    | 1.117        | CT    | 0.706        |
| 3    | VA    | 0.894        | VT    | 0.575        | RI    | 0.959        | ID    | 0.594        |
| 4    | MN    | 0.747        | PA    | 0.446        | OR    | 0.870        | MA    | 0.572        |
| 5    | NV    | 0.7107       | WA    | 0.429        | NE    | 0.533        | OR    | 0.563        |
| 6    | OR    | 0.649        | MN    | 0.402        | MA    | 0.526        | NH    | 0.560        |
| 7    | WA    | 0.598        | NE    | 0.321        | VT    | 0.372        | MD    | 0.547        |
| 8    | CT    | 0.575        | IN    | 0.273        | ND    | 0.297        | NC    | 0.496        |
| 9    | NE    | 0.481        | CT    | 0.243        | IN    | 0.217        | WI    | 0.418        |
| 10   | CO    | 0.470        | ID    | 0.196        | NC    | 0.216        | IN    | 0.375        |
| 50   | CA    | -1.329       | CA    | -1.570       | CA    | -1.370       | CA    | -1.050       |
| 49   | MS    | -1.091       | MS    | -1.050       | WV    | -1.104       | NY    | -0.807       |
| 48   | WV    | -1.090       | WV    | -0.996       | FL    | -1.003       | FL    | -0.517       |
| 47   | NY    | -1.070       | LA    | -0.862       | LA    | -0.926       | WV    | -0.469       |
| 46   | AL    | -0.736       | AL    | -0.750       | MS    | -0.760       | MO    | -0.290       |
| 45   | LA    | -0.619       | FL    | -0.704       | IL    | -0.745       | IL    | -0.283       |
| 44   | TX    | -0.443       | HI    | -0.609       | AL    | -0.712       | AK    | -0.283       |
| 43   | AR    | -0.415       | MT    | -0.540       | AR    | -0.700       | MS    | -0.232       |
| 42   | MA    | -0.323       | AR    | -0.401       | AK    | -0.670       | NJ    | -0.218       |
| 41   | OK    | -0.323       | MO    | -0.383       | NM    | -0.6107      | AL    | -0.157       |